

## Experience

### Snowflake, Senior Software Engineer

September 2022 - Present

- Spearheaded the design and development of the Sub-Database Replication feature, a key milestone for Snowflake 2.0 that reduced customer costs by 70% and increased engagement on the Snowflake Marketplace.
- Co-authored and implemented a new, scalable Replication framework that boosted developer productivity by creating a centralized architecture to automate complex replication logic, including dependency ordering and data flow.
- Designed and implemented a strategic infrastructure project to redesign the global data replication pipeline, integrating a Cloudflare R2 cache tier that slashed data transfer costs by 50-80% for public marketplace listings and enabled seamless, cost-effective cross-cloud data sharing.
- Partnered with multiple cross-functional teams to deliver critical replication logic for high-impact features like Cortex Search and Native Apps, ensuring seamless integration and functionality.
- Served as a subject matter expert for multiple critical and complex components, including DML replication which includes working with worker server in C++ to handle the data replication and encryption, Sub-Database Replication which involved optimizing graph traversal for millions of objects to calculate the transitive closure.
- Independently designed and developed a new Integration Test framework, which significantly enhanced the testing capabilities and developer experience for the Replication feature.
- Proactively identified and resolved multiple critical day-1 production issues, including a significant billing error that under-billed replication jobs by an average of 30% and complex race conditions caused by half-dead computing instances and recovery logic.
- **Technologies:** C++, Java, Python, FoundationDB, Snowflake, Jenkins

### Facebook, Software Engineer Intern

June - August 2021

- Worked on Creator Studio platform to support music labels and creators to understand their content performance in Facebook's family of apps and grow their business
- Developed data exporting feature to export metrics to CSV files using Exports (internal MapReduce framework)
- Implemented Top Country and Top Product APIs to help users understand know where their content performed best
- **Technologies:** Hack, React, Laser (internal data indexer for Hive), Saber (internal database), Exports

## Awards & Honors

ICPC World Final 2022: 44<sup>th</sup> place

C++

ICPC World Final 2021: 52<sup>nd</sup> place

C++

Vietnam Olympiad in Informatics 2017: 16<sup>st</sup> place

C++

## Projects

### Techsphere, Web Application

August 2021 - Sep 2022

- A platform providing free mentorship for college students and junior software engineers with 4000+ signed-up users
- Designed and developed core flows of the platform including Authorization, Data Models, and Scheduling
- Participated in discussion with Project Managers and Founder to plan product road map and analyze upcoming features
- Led a team of 5 engineers and utilized SCRUM model to manage and boost up performance of the team
- **Technologies:** React, Firebase Firestore, GCP Functions, GCP Hosting

## Technical Skills

**Languages:** C++, Python, Go, Java, JavaScript, SQL

**Technologies:** React, Node.js, Flask, AWS, Azure, GCP, Spark, MySQL, MongoDB, Docker, Git

**Experiences:** Web & Android App Development, Distributed Systems, Cloud Computing, Database Development

## Education

University of Rochester, Bachelor of Science in Computer Science

August 2018 - May 2022

▪ **GPA:** 3.81/4.0

▪ **Relevant Coursework:** Advanced Algorithms, Database System, Operating System, Networking